

Total No. of Questions : 10]

SEAT No. :

P1664

[Total No. of Pages : 2

[5460]-191
T.E.(Information Technology)
COMPUTER NETWORK TECHNOLOGY
(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data if necessary.*

- Q1)** a) Compare between IPv4 and IPv6. [6]
b) Explain 3 way handshaking with diagram. [4]
OR
- Q2)** a) Differentiate between POP3 and IMAP. [6]
b) Explain the concept of ARP. [4]
- Q3)** a) Explain leaky bucket algorithm. [6]
b) Differentiate between FTP and TFTP. [4]
OR
- Q4)** a) What is socket? Explain various socket primitives used in TCP. [6]
b) Explain various transport layer services. [4]
- Q5)** a) Differentiate between IEEE 802.11 and IEEE 802.16. [8]
b) Describe MAC layer mechanism of IEEE 802.11. [8]
OR
- Q6)** a) Describe Bluetooth protocol stack. [8]
b) Explain in detail CSMA/CA. [8]
- Q7)** a) What are the operating environment constraints in WSN? [8]
b) Describe each component in sensor node architecture. [8]
OR
- Q8)** a) State different MAC protocols in sensor network? Explain S-MAC in detail. [8]
b) List and explain any four applications of sensor network. [8]

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- Q9)** a) List different routing protocols used by WSN. Explain any two protocols with example. [10]
- b) Differentiate between content based & geographic routing. [8]

OR

Q10) Write short note on: (Any two): [10]

- a) i) IoT
- ii) SPIN
- iii) SDN
- b) What are different design issues and challenges of wireless sensor network? [8]

